Emergency Water Disinfection

USE ONLY WATER THAT HAS BEEN PROPERLY DISINFECTED FOR DRINKING, COOKING, MAKING ANY PREPARED DRINK, OR FOR BRUSHING TEETH

Choose disinfection method:

1. Boiling
Boiling water will kill most types of disease-causing organisms that may be present. If the water is cloudy, filter it through clean cloths or allow it to settle, and draw off the clear water for boiling. Boil water vigorously for one minute (at altitudes above one mile, boil for three minutes). Let the water cool and store it in clean containers with covers. To improve the flat taste of boiled water, aerate it by pouring it back and forth from one container to another and allowing it to stand for a few hours, or add a pinch of salt for each quart or liter of boiled water.

2. Chemical Treatment
When boiling is impractical, certain chemicals will kill most harmful or disease-causing organisms. For chemical disinfection to be effective, the water must be filtered and settled first. Chlorine and iodine are the two chemicals most commonly used to treat water. The following are methods of chemical disinfection using chlorine or iodine

- **You can use a non-scented, household chlorine bleach that contains a chlorine compound to disinfect water.** Do not use non-chlorine bleach to disinfect water. Typically, household chlorine bleaches will be 5.25% available chlorine. Follow the procedure written on the label. When the necessary procedure is not given, find the percentage of available chlorine on the label and use the information in the following table as a guide. Remember, 1/8 teaspoon and 8 drops are about the same quantity. Available Chlorine Drops per Quart/Gallon of Clear Water Drops per Liter of Clear Water 1% 0 per Quart – 40 per Gallon 10 per Liter 4-6% 2 per Quart – 8 per Gallon (1/8 teaspoon) 2 per Liter 10% 1 per Quart – 4 per Gallon 1 per Liter
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- If the strength of the bleach is unknown, add ten drops per quart or liter of filtered and settled water. Double the amount of chlorine for cloudy, murky or colored water or water that is extremely cold. Mix the treated water thoroughly and allow it to stand, preferably covered, for 30 minutes. The water should have a slight chlorine odor. If not, repeat the dosage and allow the water to stand for an additional 15 minutes. If the treated water has too strong a chlorine taste, allow the water to stand exposed to the air for a few hours or pour if from one clean container to another several times.
- **You can use granular calcium hypochlorite to disinfect water.** Add and dissolve one heaping teaspoon of high-test calcium hypochlorite (approximately ½ ounce) for each two gallons of water, or 5 milliliters (approximately 7 grams) per 7.5 liters of water. The mixture will produce a stock chlorine solution of approximately 500 milligrams per liter since the calcium hypochlorite has available chlorine equal to 70 percent of its weight. To disinfect water, add the chlorine solution in the ratio of one part chlorine solution to each 100 parts of water to be treated. This is roughly equal to adding 1 pint (16 ounces) of stock chlorine to every 12.5 gallons of water or (approximately ½ liter to 50 liters of water) to be disinfected. To remove any objectionable chlorine odor, aerate the disinfected water by pouring it back and forth from one clean container to another.
- **You can use chlorine tablets to disinfect filtered and settled water.** Chlorine tablets containing the necessary dosage for drinking water disinfection can be in a commercially prepared form. These tables are available from drug and sporting goods stores and should be used as stated in the instructions. When instructions are not available, use one tablet for each quart or liter of water to be purified.
- **You can use a tincture of iodine to disinfect filtered and settled water.** Common household iodine from the medicine chest or first aid kit may be used to disinfect water. Add fine drops of 2 percent U.S. or your country's approved Pharmacopeia tincture of iodine to each quart or liter of clear water. For cloudy water, add ten drops and let the solution stand for at least 30 minutes.
- **You can use iodine tablets to disinfect filtered and settled water.** Purchase commercially prepared iodine tablets containing the necessary dosage for drinking water disinfection at drug and sporting good stores. Use as stated in instructions. When instructions are not available, use one tablet for each quart or liter of filtered and settled water to be purified.

  Note: This information is taken and adapted from the Environmental Protection Agency (EPA).